Jinwon Lee

AI Platform Software · MLOps/Serving · System Software

Website

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This portfolio summarizes platform, infrastructure, serving, observability, and system performance work behind AI applications that need to run reliably in real operating environments. Each project is organized around the problem context, my role, design and operations focus, outcomes, and capability signals.

AI Platform Software

Production AI service operations, Kubernetes platforms, authentication and observability integration

MLOps & Serving

LLM evaluation, model-serving operations, and repeatable GitOps-based delivery flow

System Software

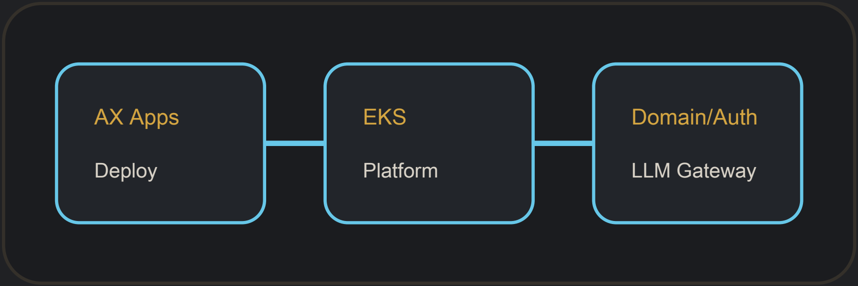
Linux runtime behavior, high-throughput packet processing, and resource-aware performance work

Cloud Infrastructure

Automation with Azure/AWS/OpenShift, Helm/Helmfile, Terraform, and Ansible

Project Details

Problem, role, design/operations focus, outcome, and capability point

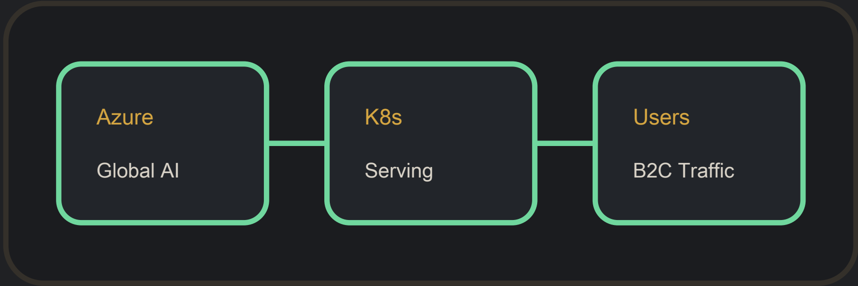


01. Internal AX Agent Platform

AWS EKS-based platform for deploying and operating AX applications with automated domains, access control, governance, and LLM Gateway integration.

Problem	Enable teams to deploy AX applications quickly while enforcing enterprise access control, security, and AI governance.
Role	Designed, built, deployed, and operated the platform while owning authentication, authorization, security, and internal governance.
Design/Ops	AWS EKS, AX Agent runtime adaptation, automated domain provisioning, access control, policy enforcement, and client integration with the existing LLM Gateway.
Outcome	Created a self-service path for governed AX application deployment with automatic domains and authorized-only access.
Capability	Experience building a governed self-service agent platform with runtime compatibility, access control, and enterprise LLM integration.

AWS · EKS · Kubernetes · AuthN/AuthZ · Governance



02. Global B2C Generative AI Service

Microsoft Azure-based B2C generative AI service expanded to the UK with initial production operations and incident response.

Problem	Expand a consumer-facing generative AI service to the UK and stabilize its initial production operations.
Role	Supported platform/SRE work for the UK rollout, initial production operations, and incident response.
Design/Ops	Azure and Kubernetes operations, rollout readiness, production stabilization, and incident response.
Outcome	Supported the UK rollout through initial stabilization and completed the engagement in June 2026.
Capability	Experience supporting UK rollout, production stabilization, and incident response for a consumer generative AI service.

Azure · Kubernetes · Generative AI · SRE

Project Details

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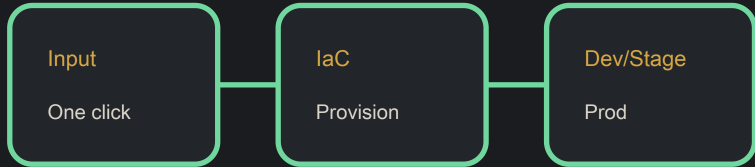
03. Company-wide AI Chatbot Platform (30K Users)

Enterprise AI chatbot platform serving 30,000 employees with Kubernetes, MLOps-style operations, and data protection.

Problem	Provide a reliable internal AI assistant platform for tens of thousands of employees.
Role	Led infrastructure, platform, and AI application operations across development, staging, and production clusters.
Design/Ops	Kubernetes operations, Helm/Helmfile IaC, AI service delivery, authentication, and data protection compliance.
Outcome	Supported a company-wide AI service with production-grade operating standards.
Capability	Experience defining Kubernetes-based AI platform operations for tens of thousands of internal users.

Kubernetes · Internal Cluster · MLOps · ArgoCD · Helm



04. Kubernetes Cluster Automation

One-click Kubernetes cluster provisioning for repeatable developer and test environments.

Problem	Cloud developers needed repeatable Kubernetes environments without manual setup overhead.
Role	Designed and implemented provisioning automation for Kubernetes clusters and shared platform modules.
Design/Ops	IaC-driven provisioning, Python automation, reusable cluster modules, and developer environment standardization.
Outcome	Reduced manual operations and improved environment consistency for cloud development workflows.
Capability	Experience standardizing infrastructure and automating repeatable development and test environments.

Kubernetes · Terraform · Ansible · Python

Project Details

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05. LLM Performance Evaluation Pipeline

MLOps-style evaluation pipeline for LLM quality and performance across Machine Translation tasks.

Problem	LLM model quality needed repeatable evaluation rather than ad hoc manual checks.
Role	Built the evaluation workflow and metric tracking path for Machine Translation model comparison.
Design/Ops	Python evaluation automation, PyTorch-based workloads, MLflow metric tracking, and Kubernetes execution.
Outcome	Made LLM evaluation results easier to compare and reproduce across model candidates.
Capability	MLOps experience automating model quality comparison and making evaluation results traceable.

Python · PyTorch · MLflow · MLOps · Kubernetes

06. GitOps CI/CD Pipeline with ArgoCD

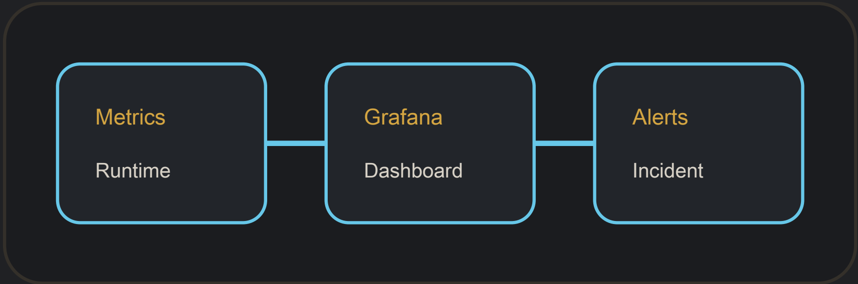
GitOps-based delivery pipelines for containerized AI/platform services with Python automation and DevSecOps checks.

Problem	Containerized AI/platform services needed repeatable, auditable, and secure deployment workflows.
Role	Designed delivery automation and connected AI service deployment pipelines with security verification steps.
Design/Ops	ArgoCD GitOps, GitHub Actions, Docker image handling, Kubernetes deployment, and Python glue automation.
Outcome	Improved deployment consistency and reduced reliance on manual release operations.
Capability	Experience connecting deployment consistency, security checks, and operations automation for containerized services.

ArgoCD · GitHub Actions · Docker · Kubernetes · Python

Project Details

Problem, role, design/operations focus, outcome, and capability point



07. Centralized Monitoring & Observability Stack

Centralized observability stack for system metrics, container runtime metrics, dashboards, and alerts.

Problem	Production services needed shared visibility into system behavior and container runtime health.
Role	Built the monitoring path and dashboards used for platform and service operations.
Design/Ops	Prometheus metrics, Grafana dashboards, OpenTelemetry signals, Loki logging, and alert rules.
Outcome	Improved operational visibility for incident detection and runtime analysis.
Capability	Experience using metrics, logs, and traces for incident detection and runtime analysis.

Grafana · Prometheus · OpenTelemetry · Loki

08. High-throughput Packet Mirroring Application

Real-time packet mirroring application that forwarded up to 130,000 packets/sec under constrained resources.

Problem	Network/application teams needed real-time packet forwarding under limited system resources.
Role	Implemented the packet mirroring application and tuned it for resource-constrained operation.
Design/Ops	High-throughput I/O, Linux runtime behavior, resource limits, and real-time data forwarding.
Outcome	Delivered a system component capable of forwarding up to 130,000 packets/sec.
Capability	Experience optimizing Linux runtime behavior and high-throughput data movement under resource constraints.

Linux · Networking · System Software · Performance

Project Details

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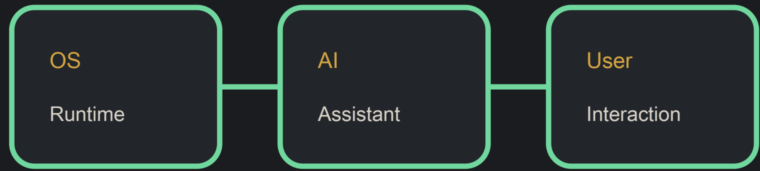
09. Distributed CNN Training System

Research on accelerated CNN distributed deep learning with automatic resource-aware layer placement.

Problem	Distributed CNN training suffered from network bottlenecks and inefficient layer placement.
Role	Contributed to research on resource-aware placement for distributed TensorFlow/CNN training.
Design/Ops	Distributed systems, model parallelism, resource-aware scheduling, TensorFlow, and GPU-aware workloads.
Outcome	Published research artifacts including USENIX NSDI 2019 poster and arXiv paper; reported up to 2.3x training speedup.
Capability	Research experience on network bottlenecks, resource placement, and performance improvement for distributed training workloads.

TensorFlow · Distributed Systems · AI Workloads · Python



10. Siri-like AI Assistant on TmaxOS

AI assistant with system-level integration and natural user experience.

Problem	The OS environment needed an AI assistant that could connect user interaction with system-level capabilities.
Role	Built AI inference and preprocessing flows and integrated them with TmaxOS-level behavior.
Design/Ops	Async architecture, system integration, AI model inference, and data preprocessing.
Outcome	Delivered an AI assistant experience with smoother interaction through asynchronous design.
Capability	Experience integrating AI functionality with OS-level behavior, asynchronous processing, and user interaction.

Python · AI/ML · TmaxOS · Async Programming

Links and References

Detailed projects and the latest CV are available on the online portfolio.

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https://jinwonlee.github.io/

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Publications

- Alleviating the Network Bottleneck for CNN Distributed Training through Automatic Resource-Aware Layer Placement — USENIX NSDI 2019 (2019)
- Accelerated Training for CNN Distributed Deep Learning through Automatic Resource-Aware Layer Placement — arXiv 2019 (2019)
- Improving Performance of Distributed TensorFlow using CNN Characteristics Exploiting Model Parallelism — ACM EuroSys 2018 (2018)

This document is a portfolio summary based on publicly shareable career and project information. It does not include confidential implementation details or non-public information.